POWER PULSE

Z-Bus Algorithm and Applications Important Formulae

Notes by Balaji Sir

1 Basics

1. $$Z_{bus} = Y_{bus}^{-1}, \qquad V_{bus} = Z_{bus}\, I_{bus}$$

   Z-bus is symmetric and, in general, full; Y-bus is sparse

2. $$Z_{ii} = \text{Thevenin impedance at bus } i, \qquad Z_{ik} = \left.\frac{V_i}{I_k}\right|_{\text{other } I=0}$$

2 Building Z-bus (branch impedance z_b added)

3. $$\text{Type 1: new bus } p \text{ to reference: } \begin{bmatrix} Z_{old} & 0 \\ 0 & z_b \end{bmatrix}$$

4. $$\text{Type 2: new bus } p \text{ to existing bus } k: \quad Z_{pi} = Z_{ki}, \quad Z_{pp} = Z_{kk} + z_b$$

5. $$\text{Type 3: existing bus } k \text{ to reference: } \quad Z_{ij}^{new} = Z_{ij} - \frac{Z_{ik} Z_{kj}}{Z_{kk} + z_b}$$

6. $$\text{Type 4: between existing buses } k, m: \quad Z_{ij}^{new} = Z_{ij} - \frac{(Z_{ik} - Z_{im})(Z_{kj} - Z_{mj})}{Z_{kk} + Z_{mm} - 2Z_{km} + z_b}$$

3 Fault analysis

7. $$I_f(k) = \frac{V_k(0)}{Z_{kk} + Z_f}$$

8. $$V_i(F) = V_i(0) - Z_{ik}\, I_f(k)$$

9. $$I_{ij} = \frac{V_i(F) - V_j(F)}{z_{ij}}$$

10. $$\text{Unsymmetrical faults: use } Z_{kk}^{(1)}, Z_{kk}^{(2)}, Z_{kk}^{(0)} \text{ in place of } Z_1, Z_2, Z_0$$